

Theoretical underpinning and empirical study on relative trade complementarity between Indo-Pacific countries

In literature, there is a theoretical underpinning on the ‘natural trading partners. Frankel, Stein and Wei (1995) argued that, in the absence of preferential trading arrangements, if a pair of countries trade relatively more with each other than with all other countries, then they can be considered as ‘natural trading partners.’ Drysdale and Garnaut (1982) had a trade complementarity index as follows;

$$C_{ij} = \sum_k \left(\frac{X_{ik}}{X_i} \right) \left(\frac{T}{T_k} \right) \left(\frac{M_{jk}}{M_j} \right) \quad (2)$$

where

K: individual commodity

X_{ik} : country i 's export of k commodity

X_i : total exports in country i .

T: total world trade

T_k : total trade of k product in the world.

M_{kj} : country j 's total import of k commodity

M_j : total import in country j .

The trade complementarity index (C_{ij}) indicates the extent to which country i 's structure of export commodities matches with countries j 's import commodities' structure. In a competitive market without trade distortion or preferential trading arrangement such as free trade agreement, trade flows between countries will dictate that if country i 's comparative advantage matches country j 's comparative disadvantage, then this pair of countries is a ‘natural trading partners.

Through mathematical derivation from the formula of C_i by Drysdale and Garnaut (1982), Yang (2023) and Fen (2024) reformulate the above C_{ij} formula as the weighted average of revealed comparative advantage (RCA) of product K in the exporting country i and the revealed comparative disadvantage of product K in the importing country j as follows:

$$RCA_{ik} = (X_{ik} / X_i) / (W_{xk} / W_x)$$

$$RCA_{jmk} = (M_{jk} / M_j) / (W_{mk} / W_m)$$

$$C_{ij} = RCA_{ik} \text{ times } RCA_{jmk}$$

$$= [(X_{ik} / X_i) / (W_{xk} / W_x) \cdot (M_{jk} / M_j) / (W_{mk} / W_m)]$$

If the C_{ij} between country i and j is high, then there is a high degree of complementarity between these two countries. Therefore, it is mutually beneficial for both of these two countries to expand their trade because they are complement with each other as the natural trading partners defined by Frankel et al (1995)..

This study investigates the relative trade complementarity of Indo-Pacific countries in the U.S. and China markets in 2018 and 2024. The data of trading commodities is based on 6-digit of HS code. The data source is from World Trade Map.

Indo-Pacific countries include Australia, ASEAN-10, India, Japan, Korea, New Zealand and Taiwan. The two export markets are China and the U.S. The purpose is to compare the relative complementarity (C_{ij}) of each Indo-Pacific countries in the two world's largest markets.

Data sources :Word Trade map.

Data needed: Time series from 2001 to 2024 or 2025 when the data is available.

1. Country i (Indo-Pacific)total export to and total import from country j (US, China, One at a time). X_i and M_j
2. Country i export to the US and China by commodities at 6-dogit HS (Harmonized System) code X_{ik}
3. Country j (US , China) import of commodity K from each country i .

4. Total world trade (export) for T
5. Total world trade (export)

This study as two parts; Part I is to analyze the overall trade commodities and Part II analyzes trade complementarity on selected product group with concentration on information and computer technology (ITC) products under HS 84 and semiconductor products under HS 8541 and HS 8542. ITC products include 8471 for computers/data processing machines, 8517 for telecommunication apparatus, and 8542 for electronic integrated circuits.

References

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